

Crude Oil Prices Forecast Based on EMD and BP Neural Network

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Abstract: The frequent fluctuations in international crude oil prices may affect the stability of the global economy and society. The fluctuation of crude oil prices has nonlinearity, uncertainty and volatility, which bring certain challenges for forecasting crude oil prices. In this paper we use hybrid model with the empirical mode decomposition (EMD) and Back Propagation Neural Network (BPNN) to predict the crude oil prices. To improve the accuracy of prediction, we firstly decompose the crude oil prices data into a series of independent intrinsic mode functions (IMFs) and residual sequences by the empirical mode decomposition method (EMD). Moreover we used BPNN to predict the Brent and WIT crude oil prices respectively. In order to show the effectiveness of the proposed method, we adopt three statistical criteria to evaluate the hybrid method. The empirical results show that EMD-BPNN has higher prediction accuracy than BPNN, the least square support vector regression (LSSVR) and EMD-LSSVR.

Key Words: Oil price forecasting, Empirical mode decomposition, Back propagation neural network, Least square support vector regression

1 Introduction

As we all know, Oil is known as “black gold” because it is the material basis for human survival and production, the most important chemical raw material and strategic resource, and an indispensable energy source material in the national economy. The oil called black gold drives the rapid development of the international oil market, and various petroleum-derived financial products appear in the investment market. Since the economic development of most developed countries and developing countries is inseparable from the supply of oil, the oil market and the financial market have both affected the fluctuation of oil prices. On the contrary, the instability of crude oil prices will also cause the global macroeconomic turmoil. In short, a little bit of turmoil in international oil prices will spread in the chain of the world economy in the form of a butterfly effect, affecting the stability of national economies. It can be imagined that the price of oil that penetrates all aspects of the national economy will greatly disturb the healthy and normal development of China's economy.

First, the correct prediction of crude oil prices can help the government to formulate more effective energy policies to a certain extent. Second, the correct prediction of crude oil prices can also help companies avoid the risk of oil price volatility, especially for oil-mining, petrochemical, and transportation industries. Third, the prediction of crude oil prices is also related to all aspects of national life. If you can accurately predict the price of crude oil, then residents can plan ahead and adjust their living plans in advance to improve their life satisfaction.

Many studies on crude oil price forecasts provide a number of technical methods that can be broadly divided into two categories, econometrics and artificial intelligence. It has been applied to econometric models of crude oil price forecasting, such as autoregressive model (AR),

autoregressive integrated moving average (ARIMA) model, vector autoregressive (VAR) model and generalized autoregressive conditional heteroskedasticity (GARCH) models [1,2,3]. Artificial intelligence techniques include support vector regression (SVR) and artificial neural networks (ANN) [4,5].

Another way to predict crude oil prices is to first try to decompose the sequence and then separately predict the subsequence after decomposition. There are many decomposition techniques, such as wavelet analysis (WT), empirical mode decomposition (EMD), singular spectrum analysis (SSA) and variational mode decomposition (VMD) [6,7,8,9]. Zhou et al [10] forecasted gold price using EMD meta-learning rate model. In order to reduce the risk of aquaculture, a new water temperature prediction method based on EMD and BPNN is proposed [11]. Compared with the traditional simulation results of the hybrid EMD-BPNN model, it is proved that capturing the non-stationary characteristics of the water temperature signal after EMD is a very effective and reliable method to predict the water temperature in intensive aquaculture.

This paper is organized as follows. Some methodologies are introduced in Section 2. Section 3 forecasts crude oil price using the proposed EMD-BPNN and compares EMD-BPNN with BPNN, the least square support vector regression (LSSVR) and EMD-LSSVR in this paper. Finally, a conclusion is given.

2 Methodology

2.1 Empirical Mode Decomposition

The EMD was proposed by Huang et al. in 1998. It decomposes the signal into intrinsic mode functions (IMF) of different frequencies by "screening" the signal. The IMF has the following characteristics: 1) maximum value and the minimum value is equal to or different from the number of zero crossings; 2) at any time, the mean of the upper and lower envelopes must be zero. The specific decomposition

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process is as follows:

Step 1: Determine the local extremum points of the data sequence $x(t)$, including local maxima and local minima.

Step 2: Find all local maxima and use the spline interpolation function to connect the upper envelope $x_{up}(t)$. Similarly, use the spline interpolation function to connect all local minima to form the lower envelope $x_{low}(t)$.

Step 3: Calculate the envelope mean $m(t)$ according to the upper and lower envelopes:

$$m(t) = [x_{up}(t) + x_{low}(t)]/2$$

Step 4: Calculate the new sequence $c(t)$:

$$c(t) = x(t) - m(t)$$

Step 5: If $c(t)$ satisfies the two characteristics of the conditional IMF, then $c(t)$ is an IMF. In this case, use $r(t) = x(t) - c(t)$ instead of $x(t)$ in step 1. If $c(t)$ is not an IMF, then replace $x(t)$ in step 1 with $c(t)$.

Step 6: Repeat steps 1 through 5 until $c(t)$ is an IMF, denoted as $c_1(t)$. The next eigenmode function can be extracted by performing the above steps on the remaining portion $r_1(t) = x(t) - c_1(t)$, where $c_1(t)$ is the first eigenmode function. This decomposition process can continue until $r_n(t)$ has only one local extremum point or becomes a monotonic function, at which point the eigenmode function has not been extracted. After the filter is finished, the data sequence $x(t)$ can be written as

$$x(t) = \sum_{i=1}^n c_i(t) + r_n(t),$$

where n is the number of eigenmode functions. $r_n(t)$ is the final residual variable, which is the main trend term of $x(t)$. $c(t)$ is the eigenmode function, and these modulo functions are almost Orthogonal.

The eigenmode functions extracted by the above decomposition have different time scales, and thus the IMF components $c(t)$ can be combined after being arranged in frequency from high to low. In practical applications, the eigenmode functions are often combined using the principle of high to low frequency. The steps are as follows:

Step 1: Calculate the mean of the eigenmode functions $c_1(t)$ to $c_n(t)$.

Step 2: Use the t test to determine the eigenmode function $c_i(t)$ whose mean value deviates significantly from zero.

Step 3: From the first eigenmode function until the $i - 1$ th eigenmode function is simply added and reconstructed into a high frequency part; the simple addition and reconstruction of the i th eigenmode function to the n th eigenmode function for the low frequency part.

2.2 Back Propagation Neural Network

The expression of the final regression function is as follows: The network consists of an input layer, one or more hidden layers, and an output layer. There is no coupling in the same layer node, and the output of each layer node only affects the output of the next layer node. The learning process of the network consists of two parts: forward and reverse. The characteristics of the back-propagation node element are usually Sigmoid:

$$f(x) = \frac{1}{1+e^{-bx}}, b > 0.$$

In the network training phase, the output data and the expected value are compared through the input layer, the hidden layer and the output layer, and if the required error degree or the number of trainings is not reached, the output layer, the hidden layer and the input layer are passed. To

adjust the weights so that the network becomes a model of adaptability. The BP network algorithm flow is as follows:

Step 1: Initialization: Set the initial values of each weight and threshold: $w_{ij}^{(l)}[0]$, $\theta_j^{(l)} (l = 0, 1, \dots, L)$ is small random numbers, generally taking a random number of 0-1.

Step 2: Input sample and expected output: Provide training samples and target output, and perform steps 3-5 for each sample. The input training samples are: $Y = (Y_1, Y_2, \dots, Y_n)$, and the target output is: $X = (X_1, X_2, \dots, X_n)$.

Step 3: Calculate the input of each layer:

$$x^{(l)} = f(s^{(l)}) = f(W^{(l)}x^{(l-1)})$$

Step 4: Calculate the training error:

$$\text{Output layer: } \delta_j^{(l)} = d_{qj} - x_j^{(l)} f'(s_j^{(l)})$$

Implicit layer and input layer:

$$\delta_j^{(l)} = f'(s_j^{(l)}) \sum(n) k = l \delta_j^{(l)} w_{kj}^{(l+1)}$$

Step 5: Correct the weights and thresholds:

$$\begin{aligned} w_{ji}^{(l)}[k+1] &= w_{ji}^{(l)}[k] + u \delta_j^{(l)} x_i^{(l-1)} \\ &\quad + \eta (w_{ji}^{(l)}[k] - w_{ji}^{(l)}[k-1]), \\ \theta_j^{(l)}[k+1] &= \theta_j^{(l)}[k] + u \delta_j^{(l)} \\ &\quad + \eta (\theta_j^{(l)}[k] - \theta_j^{(l)}[k-1]) \end{aligned}$$

Step 6: Calculating performance indicators: When all the samples in the sample set have gone through steps 3-5, that is, a training period is completed, and the error index is calculated (the error index of this paper is the mean square error):

$$E = \frac{(\sum_{q=1}^Q (E_q)^2)^{0.5}}{Q}, \text{ where } E_q = 0.5 * \sum_{j=1}^n (d_{qj} - y_{qj})^2$$

Step 7: If the error indicator meets the accuracy requirement, if $E < \varepsilon$ then the training ends, otherwise, go to step 2 and continue to the next training cycle (ε is a small positive number according to the actual situation.)

2.3 Hybrid EMD-BPNN Model

EMD can adaptively decompose non-stationary crude oil price series into several different frequencies of IMF according to its intrinsic characteristics. IMF can highlight the local characteristics of crude oil price according to its frequency recombination from high to low. The crude oil price is composed of market fluctuations, major events and trend items. The analysis of the three groups with specific economic meanings can more clearly grasp the crude oil price characteristics. At the same time, BPNN are used to model the combination of partial prediction values to obtain the final prediction value. The basic idea is to take the predicted values at the same time as input, and take the actual price as the output. After enough sample learning, establish the function mapping relationship between the predicted values of the components and the actual values; When the input variable is the predicted value of each sub-item, its output is the final predicted value of the crude oil price. In summary, the prediction method proposed in this paper can be expressed as EMD-BPNN model. The basic steps are as follows and it can be seen in Fig. 1.

The process of EMD-BPNN is as follows:

Step 1: Perform EMD decomposition on the crude oil price sequence to obtain n eigenmode functions and 1 residual component;

Step 2: Combine the eigenmode functions into two parts according to the frequency, and the remaining components are unchanged;

Step 3: Use BPNN model to perform prediction for the above three sub-sequences;

Step 4: Input the obtained predicted values of each sequence into the trained BPNN combined model to obtain the final predicted value.

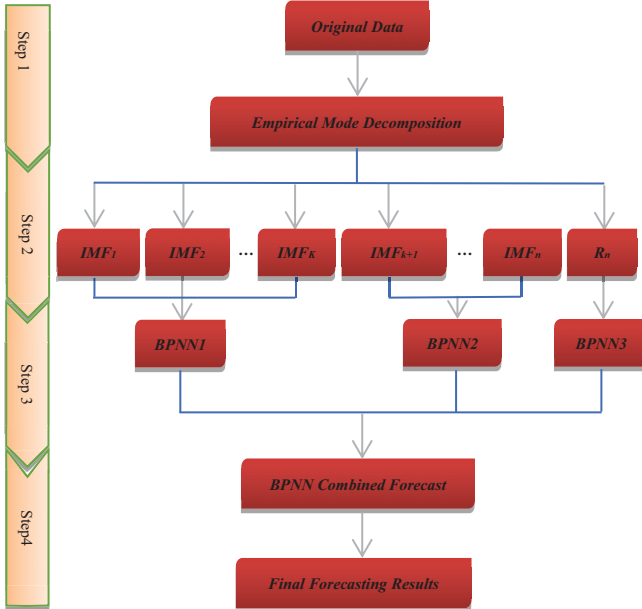


Fig. 1: Framework of EMD-BPNN learning approach

3 Empirical Analysis

3.1 Data Description and Evaluation Criteria

This paper selects WTI and Brent as experimental samples for two reasons: First, the two crude oil markets have the greatest impact on the world economy, so predictive research is useful for many countries in the world; secondly, because it is in a market Fluctuations may be an important reference for another market, so both markets are used in the empirical analysis of this paper.

This study uses daily spot prices on the US Department of Energy's (DOE) Energy Information Administration (EIA) website (<http://www.eia.gov/>). The data ranged from May 20, 1987 to June 4, 2018, excluding public holidays. For the WTI crude oil price, the number of samples is 7,831, and the data from year month to year is used for model training (observed values) and the rest as test set (observed values). The spot price of Brent crude oil, the sample size is 7878, and the difference from the WTI sample is due to the difference between the US and UK festivals. Brent crude oil price data is used for model training (observed values) from year month to year, and the rest is used as test set (observed values). The choice of data has the following two considerations: First, compared with the weekly data and the monthly data, the daily data may contain more complicated noise; according to related research, the ratio of the training set to the test set in the model construction is 4:1. Details of sample crude oil price data set are reported in Table 1.

In order to evaluate the prediction performance, this paper chooses two main criteria, namely horizontal prediction accuracy and direction prediction accuracy. Among

different error evaluation criteria, Root Mean Square Error (RMSE) and Mean absolute percentage error (MAE) have been used by scholars as effective statistical indicators to reflect the prediction level error. The calculation formulas for RMSE and MAPE are as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2},$$

$$MAPE = \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times \frac{100}{N},$$

where $y(t)$ is the true value of the crude oil price time series at time t , and $\hat{y}(t)$ is the predicted value. Obviously, horizontal accuracy is one of the most important criteria for judging the prediction effect of a model.

Another criterion is to consider the accuracy of the forecasting direction. From a business perspective, the latter indicator is even more important than the former. For business practitioners, the correct prediction of crude oil prices will help to adjust and improve business decisions in a timely manner, thereby earning more for itself. And improving the reliability of business decisions often depends on the correct prediction of the direction of crude oil prices. Based on the actual price $y(t)$ and the predicted price $\hat{y}(t)$, this paper proposes an indicator DS that predicts the accuracy of the direction of the rise and fall, which can be expressed as:

$$DS = \frac{1}{N} \sum_{l=1}^N \alpha(l) \times 100\%.$$

Here if $(y(l+1) - y(l))(\hat{y}(l+1) - y(l)) \geq 0$ then $\alpha(l) = 1$; otherwise $\alpha(l) = 0$.

Table 1: Samples of crude oil price

Oil Price	Type	Size	Date
Brent	Train data	6302	20 May 1987 - 15 March 2012
	Test data	1576	16 March 2012 - 4 June 2018
WTI	Train data	6264	20 May 1987 - 13 March 2012
	Test data	1566	14 March 2012 - 4 June 2018

3.2 Empirical Result

First, the Brent and WTI crude oil price series are decomposed by EMD [7]. Figs. 2-3 shows the decomposition result of the Brent and WTI crude oil price. Finally, 10 IMFs and 1 Res are obtained for Brent and 11 IMFs and 1 Res are obtained for WTI, in which IMF1 has the highest frequency, IMF10 has the lowest frequency, and decreases from IMF1~IMF10 frequency.

The Brent crude oil price sequence is broken down into 11 IMFs and a residual sequence, while the WTI crude oil price sequence is broken down into 10 IMFs and one residual component. Mean analysis was performed on each IMF after Brent and WTI decomposition. The result can be seen in Fig. 4.

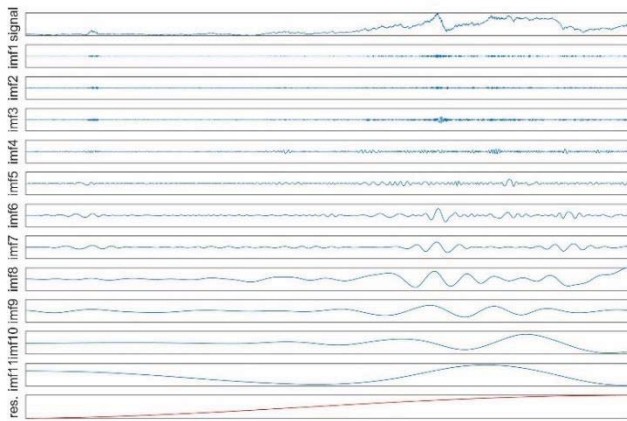


Fig. 2: EMD decomposition result of Brent

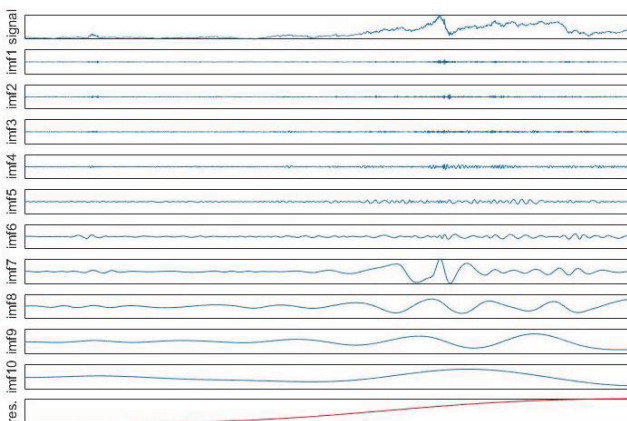


Fig. 3: CEEMDAN decomposition result of WTI

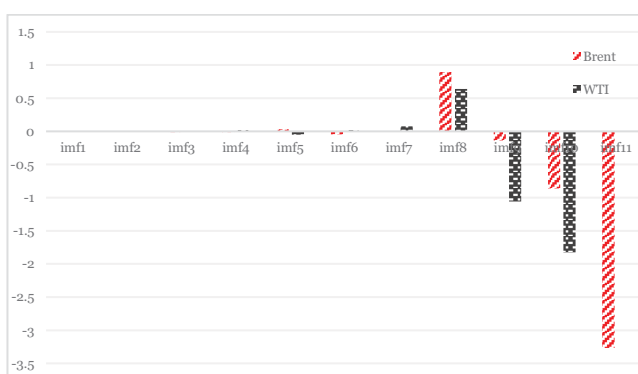


Fig. 4: Mean of IMFs of the crude oil price series

Now the average value of the IMF is tested. It is found that the first IMF whose mean value deviates significantly from 0 is IMF8, which means that IMF1~IMF7 are high-frequency components with the mean value around 0, and add them to get the price decomposition of crude oil. The high frequency part represents the normal market fluctuations of crude oil prices. The IMF8~IMF11 after

Brent decomposition is added to obtain the low-frequency part after the crude oil price is decomposed. Similarly, WTI's IMF8~IMF10 are added to represent the major event impact of the crude oil price. The remaining component remains unchanged, indicating a long-term trend term for crude oil prices. The relationship between the recombined IMF and the crude oil price series is shown in Fig. 5-6.

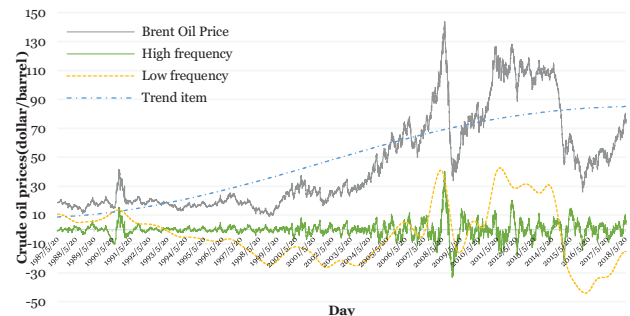


Fig. 5: Brent crude oil price series and components

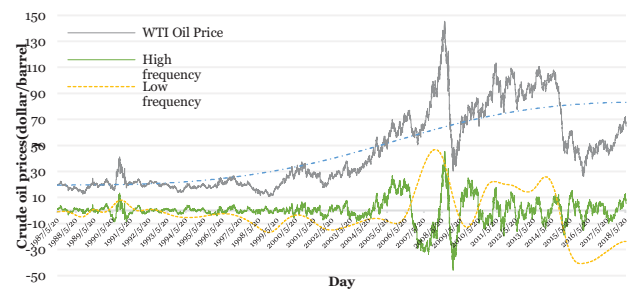


Fig. 6: WTI crude oil price series and components

As can be seen from the figure, the trend price is the main component of the oil price and has a decisive influence on the long-term trend of crude oil prices. The upward trend of the trend item and the development of the sister economy are synchronized, which also shows that the development level of the sister economy determines the long-term trend price of crude oil price. From the history of crude oil prices, although oil prices will fluctuate drastically under the influence of major events such as war, as the impact of the event disappears, oil prices will return to near the trend price. The composition of major events consists of low-frequency IMFs. It is an important part of the price of crude oil. It is subject to severe fluctuations under the influence of major events such as war. The duration is generally 4 to 5 years. In addition to being affected by the trend and the embarrassing events, the price of crude oil has also been affected by many other factors, such as weather, worker strikes and stock changes. Taking into account the impact of these factors is relatively short, these factors are classified into high-frequency market fluctuations, composed of high-frequency IMF. Although the market volatility has little effect on crude oil prices, due to the frequency of its fluctuations, market volatility is very important in the short-term forecast of crude oil prices.

After selecting the corresponding BPNN to predict each sub-item, the prediction error is shown in Table 2.

Table 2: Comparison of BPNN sub-term prediction errors

	Brent			WTI		
	MAPE	RMSE	DS	MAPE	RMSE	DS
HF	132.00	1.324	48.851	89.753	1.598	47.880
LF	0.011	0.006	99.936	0.337	0.216	99.936
T	0.019	0.027	99.936	0.014	0.014	99.936

From Table 2, the prediction ability of the BPNN model on the influence items and trend items of low-frequency major events is quite good. The performance in the high-frequency market fluctuations is very poor. The main reason is that the high-frequency market fluctuations are greatly affected by market speculation, and are also affected by factors such as weather, strikes, and inventory, which undoubtedly increase the difficulty of prediction. Combining IMFs into high-frequency and low-frequency 2 terms for prediction helps to avoid error accumulation during the prediction process and improves the final combined prediction accuracy.

The final combined prediction of WTI and Brent crude oil prices using the EMD-BPNN method is shown in Figs. 7-8.

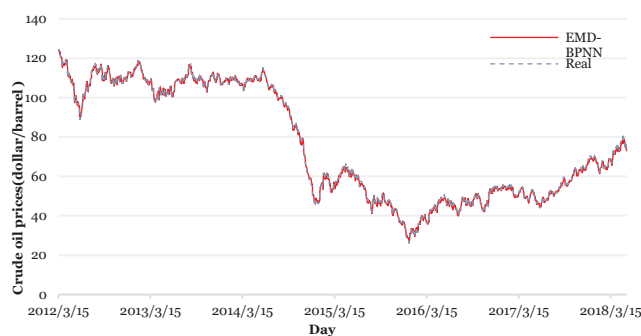


Fig. 7: Prediction results of EMD-BPNN on Brent

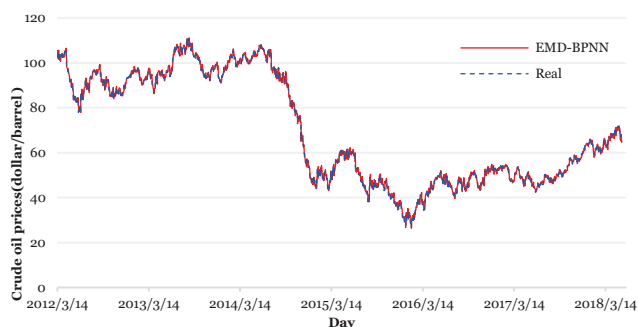


Fig. 8: Prediction results of EMD-BPNN on WTI

3.3 Comparison and Discussion

To show the superiority of the proposed EMD-BPNN, we compare this hybrid method with BPNN, the least square support vector regression (LSSVR) and EMD-LSSVR. The prediction results are shown in Table 3.

The experimental results show that BPNN is better than LSSVR for the prediction of crude oil price in single model, and EMD-BPNN is still better than EMD-LSSVR in the

hybrid model. Through the above empirical analysis of crude oil prices by BPNN and EMD-BPNN, it can be seen that whether for the RMSE or Dtat evaluation indicators, whether it is WTI or Brent crude oil price data, the accuracy of the prediction using the hybrid model EMD-BPNN is compared to the use of the nerve alone. The accuracy of the network for prediction is higher, indicating that the decomposition of the original data by the EMD algorithm can indeed improve the prediction accuracy.

Table 3: Comparison of prediction errors for different model

	Brent			WTI		
	MAP E	RMS E	DS	MAP E	RMS E	DS
LSSV R	17.18 1	14.47 9	51.90 4	17.74 5	15.60 8	51.53 3
EMD- LSSV R	17.18 1	11.23 4	51.90 4	17.74 5	15.60 8	51.53 3
BPNN	2.171	1.804	48.53 1	2.547	1.841	49.74 3
EMD- BPNN	1.510	1.273	48.66 1	1.661	1.329	50.00 8

More specifically, for WTI crude oil prices, the prediction accuracy RMSE of the EEMD-BPNN is reduced by 34.78%, the corresponding MAPE is reduced by 27.81%, the DS is increased by 0.53%. The price of Brent crude oil is lower than that of BPNN. The prediction accuracy RMSE of EEMD-BPNN for Brent is reduced by 30.47%, the MAPE is reduced by 29.43%, and the DS is increased by 0.22%.

4 Conclusion

In this paper, EMD firstly is used to decompose the non-stationary crude oil price data. Then BPNN is used to predict crude oil price. In addition, EMD-BPNN model is compared with other models such as EMD-LSSVR model, BPNN and LSSVR. The results show that EMD-BPNN is superior to EMD-LSSVR and BPNN and LSSVR in the forecast of the crude oil price.

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